

CS 6959 — Project 5

Adding Custom Instructions

a. U-encoding in TRaX versus RISC-V

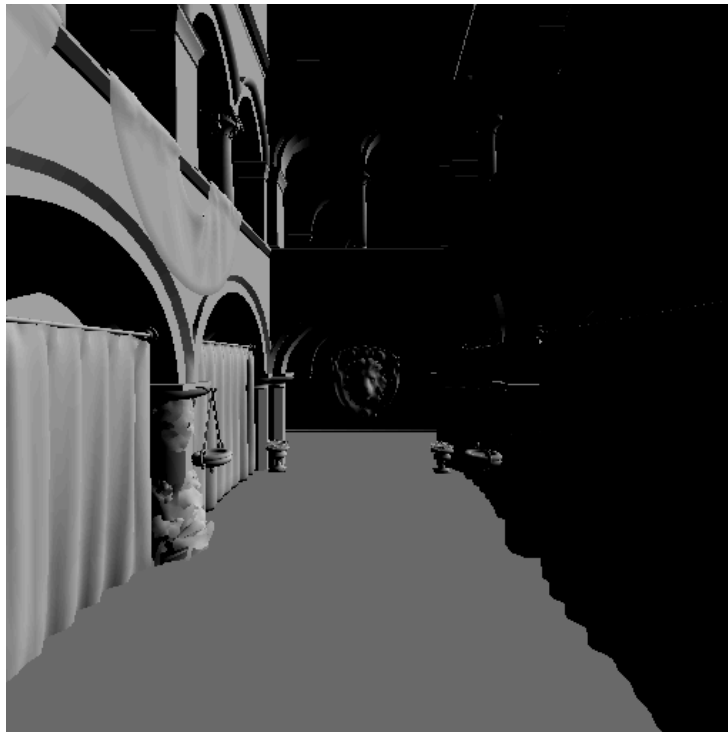
RISC-V U-format has a 20-bit immediate, a 5-bit destination register, and a 7-bit opcode. Under the custom-0 opcode, TRaX splits that immediate into `funct3 = bits[14:12]` and `func_code = bits[31:15]`, with `funct3 = 0` selecting its U-encoded instructions.

b. Why is the immediate 0x00018?

BaryInterp3 uses `func_code = 3` and `funct3 = 0`, so `imm = (3 << 3) | 0 = 0x00018`. With `rd = x0` and opcode `0xb`, the complete instruction word is `0x0001800b`.

c. Changes in kernel.dump

Without the custom instruction, interpolation uses ordinary floating-point arithmetic such as `fmul.s`, `fmadd.s`, `fadd.s`, and `fsub.s`. With it, the dump contains `0001800b .insn 4, 0x0001800b` at two compiler-generated sites, with inputs placed in `f0–f10` and results consumed from `f28–f30`.



Sponza, 512×512 , rendered using BaryInterp3.